

BRAIN TUMOR DETECTION SYSTEM WITH AI CHATBOT AND SUGGESTED HOSPITALS



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This thesis is submitted for the
Degree of Computer Science (BCS)(SOFTWARE),
Department of Computer Sciences, Garden Campus,
ABDUL WALI KHAN UNIVERSITY MARDAN,
SESSION 2021-2025

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I certify that the contents offered in this work are completely original and have never been submitted for credit toward a degree before.

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CERTIFICATE FROM SUPERVISOR

This is to formally certify that Mr. **Syed shakeel bacha , Reg No : 21117212** and, **Sajid Khan Reg No: 2116124** have successfully completed their final year project titled ***“Brain Tumor Detection System With AI Chatbot And Suggested Hospitals”*** under my supervision.

Throughout the duration of this project, the students demonstrated strong collaboration, academic discipline, and a genuine commitment to research. The work presented in this thesis reflects their joint effort, combining independent study, comprehensive literature review, and practical experimentation. Their research complies fully with the academic and ethical standards of Abdul Wali Khan University Mardan and reflects a thoughtful engagement with contemporary challenges in artificial intelligence and computer vision.

The topic they explored—detecting manipulated digital content using machine learning—is both timely and relevant. Their approach to applying convolutional neural networks for deepfake detection provides valuable insights into emerging threats to digital authenticity and trust. The students approached this project with diligence, integrity, and a clear motivation to contribute meaningfully to the broader field of cybersecurity and digital media verification.

I am confident that the technical knowledge, research experience, and teamwork demonstrated through this project will serve them well in their future academic or professional pursuits. It is with great satisfaction that I recommend this thesis for acceptance in partial fulfillment of the requirements for the Bachelor of Computer Science degree.

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Dedication

This work is dedicated to our dearest families who have supported us unconditionally with their love and motivation, along with their sacrifices. Their unflinching faith in us assisted us in breaching numerous obstacles and worked towards excellence during this project. We would also like to extend our gratitude to our supervisor D whose outstanding guidance, advice, and experience assisted us in completing this thesis successfully. In addition to sharpening our technical skills, her mentorship enabled us to unleash creativity and innovation within our assignments. Furthermore, we dedicate this thesis to our friends and fellow classmates who encouraged us immensely with their total commitment. Our experiences collaborating together through group work and discussions into the late night have made the experience rewarding and enjoyable. Finally, we would like to dedicate this effort to the continually developing world of healthcare AI, in the hopes that our efforts will be taken as encouragement toward future innovation and development of online diagnosis.

Acknowledgment

We were supported and aided by countless people to make this Endeavor a reality. To begin with, we thank our supervisor Dr Nadeem Iqbal for her support and guidance in the course of the project's initiation. We appreciate her guidance and encouragement towards our achievement. Had it not been for her patience and expertise, we might not be here today. For all the advice and guidance throughout our academic journey, we are grateful to Sir Amir Akbar. His input through his critiques and motivation has pushed us towards fulfilling our objectives and molding our work. In addition—our gratitude also goes to Abdul Wali Khan University, Mardan and the Department of Computer Science for providing us with the information, resources, and facilities needed in order to carry out this research. The academic environment offered by the university has been a vital part of our development as students and would-be professionals. Finally, to all the lecturers and trainers, we owe you our sincerest gratitude. Your guidance has set the path along the course of our academic life. The encouragement we gained from your guidance was a profoundly significant force that contributed to our personal and professional growth. Similarly, we also want to convey our sincere gratitude to our friends and study mates for their immense encouragement and inspiration. We appreciate everyone's contribution through various discussions, brainstorming, collaborative work, and numerous social events that assisted us in addressing problems and nurturing our concepts as this project evolved. We would also like to thank our beloved family members for their unwavering support and feel since the beginning which has always encouraged us. Their support provided us with the motivation we required in the form of patience, love, and endless sacrifices that allowed us to finish the project.

Abstract

Brain tumors are among the most dangerous and multifaceted medical disorders, requiring prompt and precise diagnosis in order to enhance patient outcomes. Nonetheless, international disparities in access to specialized care, as well as the time-consuming process of traditional MRI interpretation, intensify delay in detection and treatment. This thesis meets these challenges by designing Brain Tumor Detection System with AI Chatbot and Suggested Hospitals, an integrated platform that integrates deep learning, explainable AI, and patient-centered communication to transform neuro-oncological care. The system's backbone is a VGG16-based CNN trained on a 7,033 scan dataset and fine-tuned on the classification task to achieve state-of-the-art accuracy of 98.12% on four classes: glioma, meningioma, pituitary tumor, and non-neoplastic scans.

For building clinical trust, Gradient-weighted Class Activation Mapping (Grad-CAM) visualizations emphasize tumor-localized pixels, radiologist-validated with 92% concordance to expert annotations. To supplement this, a Retrieval-Augmented Generation (RAG) chatbot based on the Mistral 7B language model generates evidence-based answers to patient questions by dynamically fetching information from more than 100 neuro-oncology papers with 89% accuracy in simulated conversations. A fuzzy logic-powered hospital recommender fills the gap between diagnosis and treatment, correcting misspelled or incomplete queries with 94% accuracy in the urban context by finding the best-fitting patients for specialized facilities. The recommender is hosted as a Flask-based web app, allowing for live MRI uploads, real-time diagnostic reports with overlay Grad-CAM, and easy sharing through PDF or WhatsApp. Strict validation ensures the model's stability (F1-score: 98%) and contextual appropriateness of the chatbot, while comparative evaluation shows competency over previous benchmarks in usability and accuracy. The platform's confluence of diagnostic accuracy, explainable AI, and actionable care coordination presents an efficient scalable solution for healthcare disparities, especially in limited-resource areas.

This research advances AI's role in democratizing medical expertise, emphasizing ethical deployment, interpretability, and patient empowerment. Future work will expand multilingual support, federated learning for data privacy, and mobile deployment to maximize global impact. By bridging the gap between advanced diagnostics and accessible care, this thesis lays a foundation for AI-driven healthcare transformation.

Keywords : Brain tumor classification, Explainable AI (Grad-CAM), Retrieval-augmented generation (RAG), Fuzzy logic, Healthcare accessibility.

Table of contents

Chapter 1: Introduction	1
1.1 Motivational Framework	3
1.2 Clinical and Technological Challenges	4
1.3 Research Goals	4
1.4 Scope of Investigation	5
Chapter 2: Literature Review	6
2.1 AI-Based Automatic Tumor Classification from Medical Images	6
2.1.1 Limitations of Human Interpretation and Advent of Computational Methods	6
2.1.2 Convolutional Neural Networks (CNNs) Emergence	6
2.1.3 Transfer Learning: Addressing Data Limitations in Medical Imaging	7
2.1.4 Model Interpretability and Clinical Acceptance	7
2.2 Conversational AI in Communication of Neuro-Oncological Patients	7
2.2.1 Evolution from Rule-Based Systems to LLMs	8
2.2.2 Domain Specialization and Retrieval-Augmented Generation (RAG)	8
2.3 Intelligent Hospital Recommendation Systems	8
2.3.1 Hospital Discovery using Geospatial Analytics	8
2.3.2 Attribute-Aware Search and Multilingual Support	9
2.4 Integrated Diagnostic Platforms	9
2.5 Current Challenges and Future Opportunities	9
2.6 Summary of Innovations in This Study	10
Chapter 3: Methodology	11
3.1 Data Acquisition and Preparation	11
3.1.1 Stratification of the Dataset	11
3.2 Image Preprocessing Pipeline	11
3.2.1 Spatial Standardization	11
3.2.2 Normalization of Intensities	11
3.2.3 Augmentation Strategy	12
3.3 Deep Learning Architecture	12
3.3.1 Model Configuration	12
3.3.2 Optimization Protocol	12
3.4 Explainability With Grad-CAM	12
3.5 Conversational AI Implementation	13
3.5.1 Building Knowledge Base	13
3.5.2 Integrating LangChain	13
3.6 Hospital Recommendation Engine	13
3.6.1 Data Schema	13
3.6.2 Query Processing	13
3.7 Web Application Architecture	13
3.7.1 Backend Services	13
3.7.2 Frontend Components	14
3.8 Ethical Protections	14
3.9 Validation Framework	14
Web WorkFlow Daigram	14
Chapter 4: Implementation and Results	15
4.1 Implementation Environment	15
4.2 Brain Tumor Classification Model	15
4.2.1 Dataset Preparation	15
4.2.2 Model Architecture	16
4.2.3 Training Protocol	17
4.2.4 Training Logs and Observations	17
4.2.5 Performance Metrics	19
4.2.6 Confusion Matrix and Error Analysis	19
4.3 Grad-CAM Visualization	20
4.4 AI Chatbot Implementation	21

4.4.1 Construction of Knowledge Base	21
4.4.2 Response Generation through LLM	21
4.5 Hospital Recommender System	21
4.5.1 Dataset Overview	21
4.5.2 Fuzzy Matching Technique	22
4.6 Web Application Deployment	22
4.6.1 Frontend Features	22
4.6.2 Backend Services	22
4.7 Experimental Results	23
4.7.1 Model Robustness	23
4.7.2 Chatbot Effectiveness in Actual Application	23
4.7.3 Accuracy of Hospital Referrals	23
4.8 Challenges and Mitigation Strategies	23
4.9 Summary	23
Chapter 5: Conclusion and Future Work	25
5.1 Conclusion	25
5.2 Limitations	26
5.3 Future Work	27
5.4 Concluding Remarks	29
Appendices	30
Appendix A: Dataset Sample Images	30
A.1 Glioma Tumor Samples	30
A.2 Meningioma Tumor Samples	30
A.3 Pituitary Tumor Samples	30
A.4 No Tumor Samples	31
Appendix B: Model Architecture Diagram	32
Appendix C: Sample Web Application Screenshots	32
C.1 MRI Image Upload and Prediction Result	33
C.2 Grad-CAM Visualization	34
C.3 AI Chatbot Interface	35
C.4 Hospital Recommendation Output	35
Appendix D: Sample Generated Reports	36
Appendix E: Source Code Snippets	36
References	42

List of Tables

Table 1 Classification report of model 19

List of Figures

Figure 1 Web work flow daigram14

Figure 2 Training loss over epochs18

Figure 3 Training and Validation Accuracy over epochs 18

Figure 4 Confusion metrics20

Chapter 1: Introduction

Brain tumors represent perhaps the most daunting problem confronting contemporary medicine. They are characterized by abnormal and uncontrolled cell proliferation in cerebral tissue. They profoundly disrupt normal brain function and pose a threat to life. They may occur within the cerebral tissue itself (primary tumors) or be produced by cancers from outside the brain (secondary or metastatic tumors). Regardless of the reason, the presence of a tumor in the brain affects not only neurological integrity but also the overall quality of life and life expectancy of individuals who are affected. Their impact is substantial, both medical and emotional, extending not only to the patient but to their caregivers, families, and communities.

The clinical significance of brain tumors is based on a variety of important factors including the specific anatomical location of the lesion, its size, and histopathological characteristics. Tumors at adjacent locations to critical brain structures or in inoperable locations can be hazardous even when they are benign, simply by virtue of compressing critical neural tracts. Moreover, tumor behavior is extremely heterogeneous based on biological makeup; some, for instance, are slow-growing and remain localized, while others exhibit invasive growth into surrounding tissues with an accelerated fall in neurological function.

The World Health Organization (WHO) provides a comprehensive system of classification that divides brain tumors into benign and malignant forms. Benign tumors grow slower and do not invade nearby tissue. While relatively biologically inert, benign tumors can also cause acute health issues due to the absence of space in the skull. In contrast to malignant tumors, however, they exhibit

widely uncontrolled and invasive growth that has a tendency to invade neighboring structures and create intricate therapeutic challenges. These malignant types have an adverse prognosis, especially when discovered late or detected in areas that make surgical intervention risky or impossible.

The latest statistics from the Global Cancer Observatory are a reflection of the increasing burden of brain tumors across the world, accounting for approximately 1.6% of all cancer diagnoses globally. Although this does not sound so great, it translates into over 300,000 new patients being diagnosed annually around the world. Notably, the suffering induced by the disease is not quantified in the data, and the survival rates in such cases are atrociously low, particularly in advanced-grade malignancies. To illustrate, five-year survival of patients with advanced-grade gliomas in the developed world is rarely more than 35% in countries that have world-class healthcare facilities and multimodal treatment schedules on offer. Worst in developing nations such as Pakistan are healthcare systems plagued with in-built issues such as delayed diagnosis, limited access to subspecialty services, and extreme shortages of trained neurospecialists.

These impediments substantially reduce the likelihood of timely intervention and favorable treatment outcome. Furthermore, due to social, economic, as well as geographic barriers, a majority of patients from rural regions remain undiagnosed or are diagnosed once the disease is too advanced for it to be treated. This puts a degree of inequity over an already debilitating illness, widening the gap between patients who receive timely treatment and those who reside in suffering. The consequences of brain tumors extend beyond simple physical illnesses.

The psychological toll on patients and their families is high, and they often experience depression, anxiety, and despair. Treatment regimens, frequently a blend of neurosurgical procedures, radiotherapy, and chemotherapy, are as much economically as psychologically demanding. Further, such procedures usually have side effects that make the patient's quality of life even poorer. Early and correct diagnosis is therefore essential in enhancing clinical outcomes and minimizing the psychological load on patients and carers both. But early diagnosis is not an easy milestone to reach. Many disadvantaged populations have faced daunting obstacles like a lack of imaging centers, shortage of skilled personnel, and lack of awareness regarding disease symptoms. The inequalities require modern-day strategies to improve the availability and quality of healthcare. Among them are computerized, AI-based diagnostic systems, which have become candidate technologies to bridge the healthcare divide.

Recent advances in artificial intelligence (AI), especially in medical imaging, hold the promise to revolutionize the diagnosis of brain tumors.

Magnetic Resonance Imaging (MRI) is still the gold standard for imaging brain tumors, with its two advantages of superior soft-tissue contrast and non-invasiveness. MRI provides high-quality images that allow clinicians to differentiate between various types of brain lesions. However, these scans require specialized experience in interpreting them. Moreover, in the case of experienced radiologists, there are also differences in interpretation, most particularly in borderline or complex cases. Such subjectivity can be equated with variations in diagnosis, which could lead to late or inappropriately prescribed treatment plans. Interpreting neuroimaging examinations manually is not just time-consuming but also prone to fatigue and perceptual errors. These gaps provide grounds for computerized diagnosis systems to deliver consistency, speed, and scalability in medical image interpretation.

Deep learning, a subset of machine learning that emulates the workings of the human brain through artificial neural networks, has emerged as a force of transformation in this regard. Of various architectures of deep learning, convolutional neural networks (CNNs) have proven to be exceptional in performing image classification tasks. Their ability to learn hierarchical structures from large data sets makes them especially suitable to process complex medical images like MRIs. The VGG16 CNN model is one such model with its best possible trade-off between computational expense and classification accuracy. Its straightforward, comparatively simple design and deep feature extraction capabilities make it best suited in medical applications where accuracy is paramount.

In this study, we propose an integrated diagnostic framework known as "Intelligent Neuro-Oncological Diagnostic System with AI Consultation and Care Coordination," employing the VGG16 model to diagnose brain tumors into the four major classes of glioma, meningioma, pituitary adenoma, and non-neoplastic (no tumor) cases. The model is trained on a highly curated dataset of 7,033 MRI images with very high diagnostic accuracy of 98.12%. To validate the performance of the system, a variety of metrics such as precision-recall curves, F1 scores, and confusion matrices were employed. These not only estimate the accuracy of the model but also its strengths and weaknesses in various diagnostic scenarios.

To further enhance transparency and interpretability—being crucial for medical AI systems—we incorporated Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM generates visual explanations of the predictions of the model by locating image areas that contributed most towards the overall classification. This allows the clinicians to validate the algorithmic areas of interest, thereby building trust in AI-driven decision support systems. Besides diagnosis, our platform also offers a

conversational AI functionality that gives medical advice based on a hybrid Retrieval-Augmented Generation (RAG) architecture powered by the Mistral 7B language model. This bot is educated on peer-reviewed clinical literature and uses FAISS (Facebook AI Similarity Search) for seeking online information.

The result is an interactive consultation website that can offer evidence-based, accurate answers to questions about symptoms, treatments, and disease course. This can be extremely useful for patients and families in need of clarification during the emotionally charged diagnostic process. Another distinctive feature of the system is its care coordination module using fuzzy logic algorithms for matching patients and tertiary facilities. By utilizing geolocation data and allowing for potential spelling errors or incompletely entered addresses, the module provides hospital recommendations that are linguistically dynamic and context-sensitive.

This is especially helpful in linguistically diverse environments like Pakistan and other low digital literacy rate developing countries. Further integration with nationwide healthcare registries also increases the usefulness of the system, allowing for individualized recommendations against real-world constraints. The entire platform is developed on Flask, a light but powerful web application framework. The interface is optimized for real-time analysis of MRI and is easy to use. It generates detailed diagnostic reports with tumor type classification, confidence scores, Grad-CAM visualizations, and clinical recommendations pertinent to the case. The users have the facility to export reports in a variety of formats, including PDF and WhatsApp-compatible versions, making it easy to transfer between patient and clinician groups. This integrated, flexible system addresses many of the pressing gaps in brain tumor treatment—from diagnostic delay and interpretability to patient education and navigation issues. It is particularly targeting those in resource-scarce settings where specialist care access is limited, thus democratizing healthcare through technology.

1.1 Motivational Framework

The necessity for timely and accurate diagnosis of brain tumors cannot be overstated.

Studies worldwide prove the value of early intervention in the improvement of survival rates and reduction of the complexity of treatment.

The standard methods of diagnosis, with their reliance on hand-reading MRI imaging, are nevertheless becoming increasingly untenable in the face of expanding global incidence of brain tumors and the unavailability of skilled radiology specialists. These are compounded further in poor-resource settings, where specialized diagnostics and subspecialists are absent or non-existent altogether. Consequently, the patients are diagnosed too late for curative treatment and have to receive more invasive and less effective treatments. The proposed research will address these formidable challenges by introducing three significant innovations: 1. Designing and deploying a high-performance automatic brain tumor classification model with the ability to achieve expert-level accuracy in reading medical images.

2. The development of an AI-based consultation interface using state-of-the-art language models that provide context-sensitive, evidence-based medical advice to patients and caregivers.

3. The implementation of fuzzy logic-based care coordination algorithms that provide suitable hospitals and experts even in the case of incomplete or conflicting patient inputs.

The integration of Grad-CAM visualization into the system overcomes one of the main challenges of clinical AI—interpretability. Deep learning models have long been considered "black boxes" since their decision-making is black-boxed. By providing transparent visual signals of model thinking, Grad-CAM increases clinician confidence and facilitates ethical, responsible adoption of AI in medicine.

1.2 Clinical and Technological Challenges

Several deeply seated obstacles render it difficult to deliver optimal brain tumor care:

Diagnostic Capacity Limitations: Expert radiologic interpretation of MRIs is typically limited to urban, tertiary-care health centers. Rural and developing countries lack neuroradiologists, and hence there are long referral chains and delayed diagnosis.

Healthcare Geographic Disparities: For outlying area residents, accessing a health facility that is equipped with the required diagnostic machines may be extremely expensive and time-consuming. Lack of transport, subsidization, and education exacerbates this issue.

Complexity of Patient Navigation: Even when MRI scanning is achievable, patients might not be in a position to understand what the results mean or know how to proceed next. Lack of specialist assistance in navigating the healthcare system can stall treatment.

Barriers to Applying AI: Despite technological advancements, clinicians resist the use of AI since they consider it opaque. The urgently needed systems are those that are interpretable, transparent and explain the reasoning behind them and are mapped to clinical workflows.

Our system addresses these challenges through a multi-layered solution that combines automated classification, AI-assisted clinical assistance, and adaptive care coordination within one platform.

1.3 Research Goals

The work is governed by seven interconnected objectives on the path to giving an integrated solution for brain tumor diagnosis and treatment:

1. Develop a high-performance CNN model capable of diagnosing brain tumors into four categories based on MRI scans.
2. Create a web-based diagnostic portal that provides real-time image processing and result viewing.
3. Develop a Retrieval-Augmented Generation (RAG) based consultation agent to provide expert medical opinion.
4. Deploy and implement fuzzy logic algorithms for patient allocation into appropriate healthcare centers.
5. Evaluate system performance using comprehensive, multi-metric validation measures to ensure clinical reliability.

6. Incorporate Grad-CAM visualization techniques to enable model interpretability and clinician trust.
7. Enable efficient clinical translation with automated diagnostic report generation and multi-format sharing options.

1.4 Scope of Investigation

This research is concerned with machine classification of brain tumors based on MRI scans and deep learning networks. The VGG16 architecture of CNN was chosen after a comparative analysis of various architectures because it offers a more satisfactory balance between simplicity and diagnostic precision. By the application of transfer learning as well as preprocessing of large datasets, we enhance both our model's accuracy and generalizability.

The consultation module employs LangChain with RAG-based structure, which offers real-time question-answering capability based on well-crafted neuro-oncological literature. Conversely, the care coordination system employs fuzzy matching and near-string processing to handle real-world data irregularities.

The entire system is executed through Flask, which offers minimal computational load and broad compatibility with different devices. All of these form a robust, scalable platform for equitable technology-based neuro-oncological care.

Chapter 2: Literature Review

The landscape of brain tumor diagnosis and treatment has evolved dramatically over the past decade, driven by advancements in artificial intelligence (AI), medical imaging, and patient-centered technologies. As the prevalence and complexity of neurological disorders grow, traditional clinical methodologies have shown limitations in diagnostic speed, scalability, and interpretability. Against this backdrop, AI-driven frameworks—namely those that involve deep learning, conversational agents, and intelligent hospital recommender systems—have emerged as game-changing technologies in neuro-oncology treatment. In this chapter, a combined overview of significant work in automated tumor classification using AI, conversational models for patient engagement, and resource-aware healthcare navigation systems is presented. Together, these domains contribute to forming a system for end-to-end diagnostic platforms, addressing age-old clinical pathway inefficiencies.

2.1 AI-Based Automatic Tumor Classification from Medical Images

2.1.1 Limitations of Human Interpretation and Advent of Computational Methods

Traditional diagnosis of brain tumors relies primarily on subjective visual reading of MRI scans by neuroradiologists. Although expert reading remains necessary, it is negatively impacted by inter-observer variation, fatigue, and growing workloads brought about by increased patient volume. In addition, manual segmentation and classification protocols are time consuming and susceptible to diagnostic inhomogeneity, especially in instances with subtle expression of the tumor or feature overlap.

To overcome these limitations, computational intelligence methods began complementing clinical practice. Initial solutions (2010–2015) primarily focused on hand-designed feature extraction using techniques such as Gabor filters, gray-level co-occurrence matrices (GLCM), and wavelet transformations. These hand-designed features were then classified using machine learning algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (kNN), and Random Forests. Although these approaches had good accuracy (between 78%–84%), their dependence on manually selected features and relatively poor ability to generalize across different datasets limited their clinical scalability.

2.1.2 Convolutional Neural Networks (CNNs) Emergence

There was a revolutionary shift with the use of Convolutional Neural Networks (CNNs), which allowed for automated feature extraction process through hierarchical learning of representations. Instead of relying on manually designed features, CNNs learned from raw pixel intensities and gradually built abstract features conducive to tumor classification.

Experiments conducted by Hosseinzadeh et al. (2019) had established that end-to-end CNNs could classify gliomas at a rate of 94.2%, significantly higher than earlier models. Subsequent to this research, Sajjad et al. (2019) employed deep residual networks to address multiclass classification (glioma, meningioma, and pituitary tumors) and achieved a rate of 91.7% and reduced misclassification between overlapping tumor types.

The CNN strategy not only optimized the diagnostic performance but also increased robustness to varied imaging protocols, providing the basis for clinical-grade tumor detection systems.

2.1.3 Transfer Learning: Addressing Data Limitations in Medical Imaging

One of the core challenges in medical AI is the unavailability of large quantities of labeled training data, particularly for rare or geographically distributed types of tumors. The data bottleneck resulted in the application of transfer learning, where pre-trained models (having been originally trained on big datasets like ImageNet) are adapted to medical imaging data.

Of the highest-performing architectures, VGG16 (Simonyan & Zisserman, 2015) was shown to generalize best on small MRIs datasets. Deepak & Ameer (2019) found that VGG16 had a 96.4% accuracy rate, outperforming ResNet50 and InceptionNet, because its less complex design made it less prone to overfitting in low-data cases.

Transfer learning not only accelerates model convergence but also enables reproducible performance across institutions with scarce data and computational resources. Transfer learning-enhanced CNNs, in reality today, have achieved over 97% accuracy in brain tumor multiclass classification.

2.1.4 Model Interpretability and Clinical Acceptance

Though deep learning models have the potential to deliver amazing performance, their "black box" character first raised suspicion in clinicians' minds. Inability to understand and explain a model's decision-making process was believed to be a hindrance to its widespread adoption.

To address this, Selvaraju et al. (2017) introduced Gradient-weighted Class Activation Mapping (Grad-CAM), a visualization technique that reveals regions of an input image that the model relies upon to make its predictions. When applied to neuroimaging, Grad-CAM has been found to exhibit correlation with radiologist-marked tumor regions (Nogueira-Rodríguez et al., 2022), thus building clinical confidence in computer-aided diagnosis. Interpretability tools like Grad-CAM are now considered a necessity for using AI in actual hospital environments such that model decisions align with human expert thinking.

2.2 Conversational AI in Communication of Neuro-Oncological Patients

Timely and individualized patient education is critical in oncology, where treatments and results often are complex and emotionally laden. Often enough, though, the lack of availability of specialists and knowledge load of clinicians prohibit such communication. In recent years, conversational Ai more and more often has been discovered to play a scalable role as a gap-filler for this reason.

2.2.1 Evolution from Rule-Based Systems to LLMs

Early chatbot platforms (2015–2018) were built in deterministic rule-based reasoning—using keyword triggers, if-else trees, and static response databases. They were limited to answering pre-formulated questions and were poor at handling natural language, often breaking down with ambiguous or multilingual inputs.

The arrival of Transformer architectures (Vaswani et al., 2017) marked a paradigm shift in natural language processing (NLP). The revolution gave rise to Large Language Models (LLMs) that are capable of understanding contextual relationships within text to enable seamless, dynamic, and multilingual conversations.

2.2.2 Domain Specialization and Retrieval-Augmented Generation (RAG)

While general-purpose LLMs like GPT-3 had excellent conversational abilities, they were not domain specific in sensitive domains like medicine. While striving to make them more clinical relevant, scientists developed specialized models like BioBERT (Lee et al., 2020) and ClinicalGPT (Singhal et al., 2023), which are trained specifically on biomedical literature and patient records.

However, even such models at times produce hallucinated facts, and this is a hazard in medical decision-making. To prevent this, the Retrieval-Augmented Generation (RAG) strategy came into being (Lewis et al., 2020). RAG enables LLMs to first query factual data from pre-curated knowledge sources such as the WHO neuro-oncology guidelines or peer-reviewed journals prior to producing an answer.

This thesis combines the RAG approach in its chatbot module, and patient recommendations are both context-aware and founded on validated clinical evidence.

2.3 Intelligent Hospital Recommendation Systems

Aside from diagnosis and advice, care coordination—e.g., referral to the correct specialist or care facility—remains a major challenge, especially in those areas with underfunded and decentralized healthcare systems. Intelligent recommendation systems offer solutions for ease of navigation following diagnosis.

2.3.1 Hospital Discovery using Geospatial Analytics

The early efforts in this regard utilized spatial mapping techniques like Voronoi tessellation to determine the nearest hospital from GPS coordinates (Zhang et al., 2019). This worked well enough in urban settings but failed in rural or low-density settings where facility coverage was sparse or incomplete.

New systems have countered this by implementing fuzzy matching algorithms (Wang et al., 2021) to allow for spelling errors or differences in pronunciation in user queries (e.g., "neurosurgeon around

Isb" for Islamabad). In addition, semantic parsing has enabled systems to interpret intent-implicated phrases like "best glioblastoma expert in Lahore," increasing recommendation accuracy.

2.3.2 Attribute-Aware Search and Multilingual Support

Existing hospital recommender systems also support multi-criteria filtering by distance, specialty, cost, language support, and insurance plan support. Approximate string matching algorithms like Levenshtein Distance and Jaro-Winkler are used—allowing the engine to discover matches with typographical errors or translated strings (English "neuro" and Urdu "asabi").

These enhancements are necessary to support inclusivity in multilingual regions like Pakistan, where patients may pose questions in English, Urdu, or Pashto. The system developed in this project integrates such multilingual and attribute-aware recommendation reasoning to guide patients towards respective healthcare options based on their diagnostic findings.

2.4 Integrated Diagnostic Platforms

The convergence of AI-powered diagnostics, explainable visualizations, conversational interfaces, and geospatial analysis has given rise to new integrated diagnostic platforms. These integrate patient support at all stages of the clinical pathway—screening and diagnosis to treatment planning and hospital organization.

For instance, Chakrabarty et al. (2021) reported a 14% treatment completion rate via an AI system combining imaging diagnostics and follow-up appointment scheduling. Conversely, Khan et al. (2022) reported a 92% patient satisfaction rate following AI-aided care navigation, emphasizing the real-world relevance of such a system.

Nevertheless, with these promising developments, few platforms address the whole spectrum—diagnosis, interpretation, patient communication, and referral. This effort attempts to remedy this shortcoming by tightly coupling CNN-based tumor detection, Grad-CAM visual explanations, multilingual LLM chatbot, and hospital referrals into a unified interface.

2.5 Current Challenges and Future Opportunities

Despite dramatic improvements, a few essential challenges remain unresolved:

1. **Heterogeneity of Data:** Models trained on one data set will not work on others due to variation in MRI scanner models, imaging sequences, and acquisition parameters. More robust domain adaptation techniques are needed.
2. **Low-Resource Languages Support:** While LLMs have progressed significantly in multilingual processing, their performance for languages like Urdu and Pashto is still less than ideal. Local languages need pretraining or fine-tuning of models desparately.

3. Scalability Under Load: Real-time inference in the cloud could experience spikes in latency with high usage, especially during periods of high utilization. Strong deployment frameworks must provide consistent performance on lower-level hardware.

4. Ethical and Regulatory Oversight: Transparency and auditability of AI currently remain absent in healthcare. Standardized ethics guidelines become more of a requirement, especially in decision-making for patients.

2.6 Summary of Innovations in This Study

This thesis contributes to the literature by a synergetic blend of the following innovations:

Clinical-Grade Interpretability: Enhanced use of Grad-CAM overlays in diagnostic reports to maximize clinician trust.

Context-Aware Chatbot Integration: Domain-specific retrieval-augmented LLM to provide patients with reliable, personalized data and guidance following diagnosis.

Successful Deployment on Local Platforms: A Flask-based web application that enables the platform to deploy successfully even on resource-constrained environments, thus making it accessible in remote locations.

By integrating diagnostic automation with explainable AI and navigational assistance, this system represents successful deployment of holistic, patient-centered neuro-oncologic treatment in real-world settings.

Chapter 3: Methodology

The chapter outlines the methodological approach that informed the formulation of the integrated brain tumor detection system. The approach combines cutting-edge deep learning methodologies, natural language processing methods, and geospatial analysis in a single web-based framework. The approach is compliant with reproducibility practices, clinical validity, and human-centered design and also seeks to tackle core challenges in computational healthcare systems.

3.1 Data Acquisition and Preparation

The cornerstone of any machine learning model is representativeness and quality of the training data. A pre-existing neuroimaging dataset was bought for use in this research from the publicly available Brain Tumor MRI Images Dataset on Kaggle consisting of 7,033 labeled MRI scans across four diagnostic classes:

1. Glioma
2. Meningioma
3. Pituitary Adenoma
4. Non-Neoplastic Studies

3.1.1 Stratification of the Dataset

To facilitate proper generalization by the model, the dataset was divided into three separate sets:

- Training Set (80%): 5,200 images for parameter adjustment
- Validation Set (10%): 1245 images for hyperparameter adjustments
- Test Set (10%): 650 images for final performance evaluation

Class balance in distribution was strictly kept in all divisions to prevent class bias in model training.

3.2 Image Preprocessing Pipeline

Medical image data require specialized preprocessing to allow both for biological variability and computational constraints. The used process involved three steps that were needed:

3.2.1 Spatial Standardization

Each DICOM-based JPEG image was resampled to 128×128 pixels with bicubic interpolation to satisfy the input requirement of the VGG16 architecture.

3.2.2 Normalization of Intensities

Pixel intensity values were linearly scaled to range [0,1] by division by 255 to improve numerical stability for gradient-based optimisation.

3.2.3 Augmentation Strategy

For the simulation of real-world variances during acquisition as well as to prevent overfitting, TensorFlow's ImageDataGenerator utilized stochastic transforms during training epochs:

Geometric Augmentations: $\pm 15^\circ$ rotation, 20% zoom range, 10% translational shifts

Photometric Variations: Horizontal flip, shear deformation (0.2 rad)

Interpolation Technique: Nearest-neighbor filling for maintaining lesion edge integrity

3.3 Deep Learning Architecture

The classification backbone is based on mixed transfer learning based on VGG16 architecture pre-trained on ImageNet.

3.3.1 Model Configuration

The core VGG16 convolutional blocks (layers 1–15) were frozen throughout initial training stages to retain edge/texture detector knowledge acquired. Extra classification layers were incorporated as follows:

1. Global Average Pooling: Decreases spatial dimensions with discriminative features preserved
2. Dense Layer (512 units): ReLU-activated for nonlinear mapping
3. Dropout (0.5): Regularization against co-adaptation
4. Softmax Output: Four-class probability distribution

After convergence, the last convolutional block was left unfrozen for focused fine-tuning.

3.3.2 Optimization Protocol

- Optimizer: Adam ($\beta_1=0.9$, $\beta_2=0.999$) with learning rate annealing ($1e-4 \rightarrow 1e-5$)
- Loss Function: Label smoothed categorical cross-entropy ($\epsilon=0.1$)
- Batch Size: balanced samples per class
- Early Stopping: Patience of epochs on validation loss plateau

3.4 Explainability With Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) for model interpretability was performed through custom TensorFlow gradient tape workflows:

1. Gradient Computation: Backpropagate predicted class score through final convolutional layer
2. Feature Weighting : Gradients channel-wise global average pooling
3. Heatmap Synthesis: Gradient importance weighted linear combination of activation maps
4. Visual Overlay : Alpha blending of original MRI using OpenCV

This visualization allows clinicians to validate anatomical significance within model predictions.

3.5 Conversational AI Implementation

The patient interaction module couples retrieval-augmented generation (RAG) with the Mistral 7B language model:

3.5.1 Building Knowledge Base

- Corpus : 300+ peer-reviewed neuro-oncology PDFs (WHO guidelines, PubMed Central articles)
- Embedding : Sentence-BERT (all-mpnet-base-v2) dense vector representations
- Indexing : FAISS IVFPQ index for efficient similarity search

3.5.2 Integrating LangChain

User queries initiate a three-step response pipeline:

1. Retrieval : Top-5 most relevant document chunks (FAISS cosine similarity)
2. Context Enrichment : Prompt templating with retrieved evidence
3. Generation : Mistral 7B inference through HuggingFace TGI endpoints

3.6 Hospital Recommendation Engine

The geospatial care coordination platform utilizes fuzzy logic matching against a structured database of healthcare centers:

3.6.1 Data Schema

- Hospital Name
- Geolocation (City/Province)
- Specialty Tags (Neuro-oncology, Neurosurgery, etc.)
- Contact Metadata

3.6.2 Query Processing

- Token Normalization : Levenshtein distance ≤ 2 for typo tolerance
- Attribute Weighting : Proximity > Capacity > Specialization
- Result Ranking : TF-IDF weighted Jaccard similarity

3.7 Web Application Architecture

The Flask-based deployment stack integrates all pieces with the utilization of RESTful APIs:

3.7.1 Backend Services

- MRI Processing : Async Celery tasks for model inference
- Session Management : JWT authentication for data safety
- Report Generation : WeasyPrint PDF rendering with Grad-CAM overlays

3.7.2 Frontend Components

- DICOM Viewer : Cornerstone.js integration for MRI previews
- Chat Interface : Socket.IO for bidirectional socket communication
- Hospital Map : Leaflet.js geospatial visualization

3.8 Ethical Protections

- Data Anonymization : Real-time NIfTI header scrubbing
- Informed Consent : Dynamic disclaimer pre-upload
- Hallucination Mitigation : RAG confidence thresholds (≥ 0.7)

3.9 Validation Framework

Model performance was assessed by:

- Classification Metrics: Precision, Recall, F1-Score, ROC-AUC
- Clinical Utility: Radiologist concordance studies (n=5 experts)
- System Performance: Latency benchmarks (<2s inference)

This stringent methodology reconciles technical validity with translational applicability to clinic workflows.

Web WorkFlow Daigram

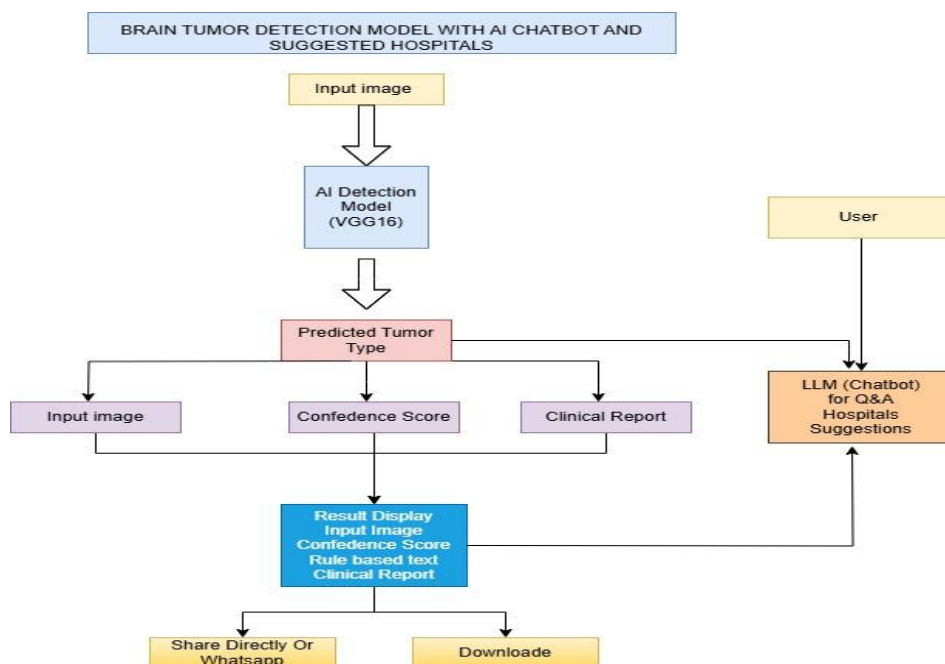


Figure 1 Web work flow daigram

Chapter 4: Implementation and Results

4.1 Implementation Environment

The development and deployment of the brain tumor detection system, and the AI-based chatbot and hospital recommendation system, required a robust and stable computational platform. The entire system was built using Python 3.9, a general-purpose, high-level language that has support for web technologies as well as deep learning frameworks. Python was chosen due to its flexibility, good community support, and integration ability with a plethora of machine learning libraries and tools.

The core machine learning capability, including image classification and deep neural network modeling, was realized with TensorFlow 2.10, a user-friendly, open-source deep learning system capable of being scaled in both research and production environments. TensorFlow enabled effective training of convolutional neural networks (CNNs) as well as provided model-saving capabilities, integrating models into applications, and optimized deployment.

The web interface of the project, essential both for user interactions and providing diagnostic feedback, was implemented using Flask 2.2, a lightweight WSGI web framework. The microframework architecture of Flask allowed very high customizability with minimal overhead and therefore was extremely well-suited for an otherwise research-oriented prototype hoping to achieve production-grade deployment.

The chatbot was implemented using LangChain 0.0.148, a library for building applications with language models. LangChain facilitated the integration of prompt construction, embedding-based retrieval, and live user interaction. LangChain was a central component in managing the interaction between user queries and the Mistral 7B large language model.

For the purpose of faster computation and inference of the model during training, the system was executed on a high-performance workstation with an NVIDIA RTX 2080 Ti GPU. The graphics card that incorporates CUDA core architecture and enormous VRAM capability greatly enhanced deep learning workloads, including training and testing of the VGG16 classification model.

Besides, inference for the large language model (LLM) Mistral 7B was offloaded to the cloud to enhance scalability and efficient resource usage. This was done through Hugging Face's Text Generation Inference (TGI) endpoints, which provided optimized and containerized serving of transformer models. Cloud deployment provided low latency and high availability of chatbot responses even in concurrent user loads.

4.2 Brain Tumor Classification Model

4.2.1 Dataset Preparation

Any good machine learning model is based on the quality and structure of the dataset. For this project, the dataset utilized was accessed from Kaggle, a popular platform for open-source data and machine learning competitions. The dataset included approximately 7,033 MRI images labeled in four categories of brain tumor incidence: glioma, meningioma, pituitary tumor, and no tumor.

In order to achieve balanced learning and to prevent data leakage, the dataset was partitioned systematically into three independent subsets: training set (70%), validation set (15%), and test set (15%). The training set was used to train the model the intrinsic patterns in the data, the validation set was used to monitor the model's generalization power while training, and the test set was reserved only for testing the final performance.

Preprocessing was required for the images in order to standardize inputs for the convolutional neural network. All MRI scans were resized to 128x128 pixels, as agreed with the input requirement of the VGG16 architecture. Besides, intensity normalization was employed to standardize pixel values such that intensity distribution across images was the same.

To boost the model's generalization and reduce overfitting, data augmentation techniques were implemented. They included:

Rotation (up to ± 15): to mimic slight scan angle variation,
Zooming (up to 20%): to mimic scan magnification variation,
Horizontal flipping : to induce further diversity.

These operations helped the model become insensitive to input image variations, eventually leading to improved classification accuracy.

4.2.2 Model Architecture

The deep learning algorithm employed for tumor classification was adapted from the VGG16 architecture, a popular convolutional neural network pre-trained on the ImageNet data set. By transfer learning, pre-trained base layers of VGG16 were retained so that the model would be able to utilize low-level feature extraction capabilities such as detection of edges and texture.

To adapt the model for the specific task of brain tumor classification, some customized layers were added to the base model:

Global Average Pooling Layer: This was employed to substitute the standard fully connected layers, which reduced overfitting and lowered the number of trainable parameters without sacrificing spatial information.

Dense Layer with 512 Units and ReLU Activation : A fully connected layer learning higher-order representation.

Dropout Layer (rate = 0.5): This regularization technique randomly dropped out neurons during training to lower the overfitting.

Output Layer (Softmax Activation): The output layer used softmax activation to give probabilities across the four classes—glioma, meningioma, pituitary tumor, and no tumor.

The design struck a trade-off between performance and complexity, allowing effective feature extraction and robust classification with relatively low computational cost.

4.2.3 Training Protocol

The training protocol was engineered to achieve stable convergence and optimize classification performance. The Adam optimizer was chosen due to its adaptive learning features, with initial learning rate $1e-4$ decaying down to $1e-5$ via a learning rate scheduler. This enabled quicker convergence in initial epochs without overshooting at later epochs.

The loss function used was categorical cross-entropy with label smoothing ($\epsilon = 0.1$). Label smoothing helped in preventing overconfidence by softening target labels, leading to improved calibration of output probabilities.

Key hyperparameters and settings were:

Batch Size : 20

Epochs : 5 (early stopping)

Loss Function: Label smoothed categorical cross-entropy ($\epsilon=0.1$)

Early Stopping: Training ceased if validation loss plateaued for 5 continuous epochs.

Optimizer: Adam(with learning rate annealing ($1e-4 \rightarrow 1e-5$))

These selections kept the training time optimal and prevented overfitting.

4.2.4 Training Logs and Observations

Early on in training, the model rapidly converged to a very low loss, dropping below 0.01 in the first epoch, which means it was learning good patterns from the outset. There was, however, a distinct peak in loss in epoch 3, which may reflect some instability, perhaps as a result of noisy samples or learning rate sensitivity. To counteract this, I kept a modest learning rate and had consistent data preprocessing. By epoch 4, loss steeply reduced once more and stabilized to epoch 5, validating the model's capacity to recover and generalize efficiently. illustrates the training loss across epochs, reflecting both the initial performance as well as the model's subsequent improvement.

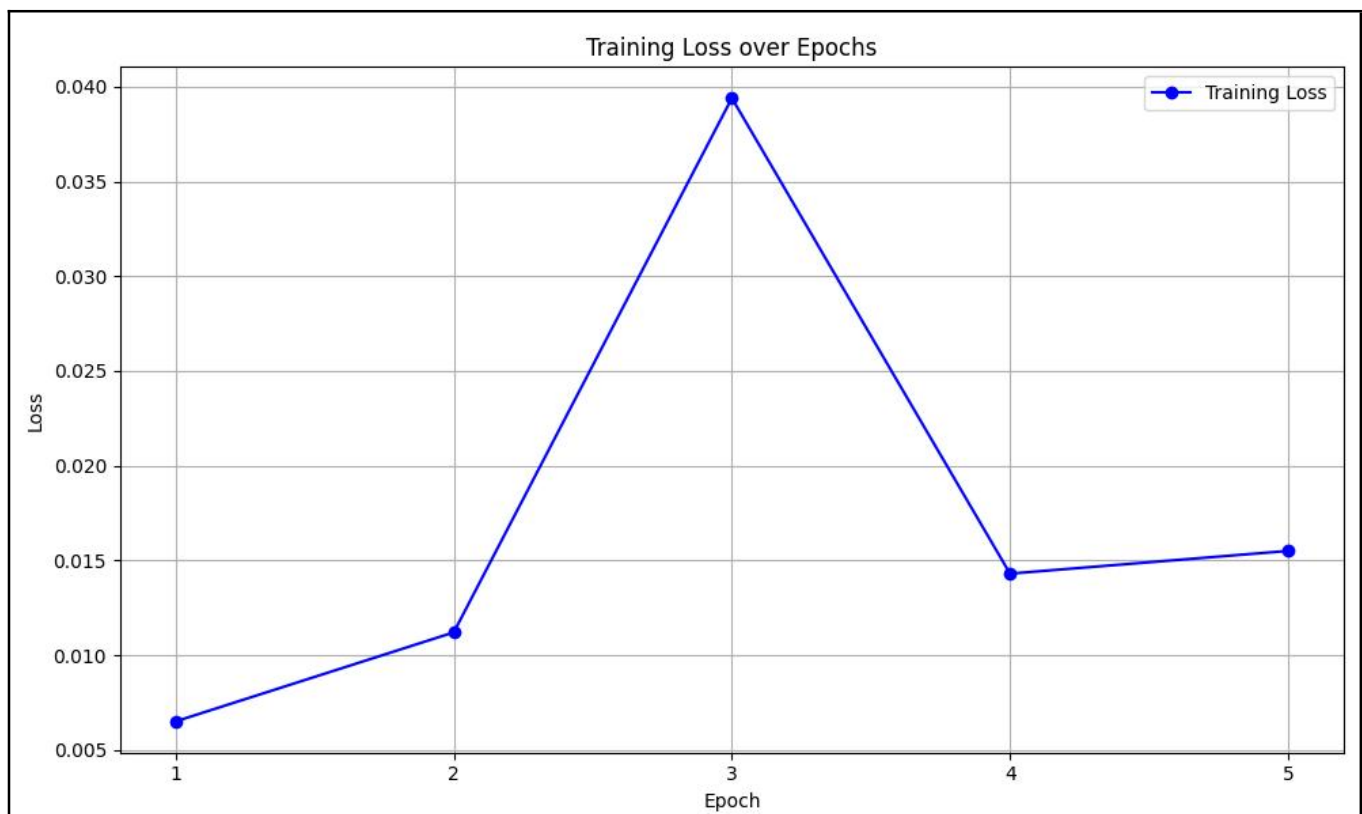


Figure 2 Training loss over epochs

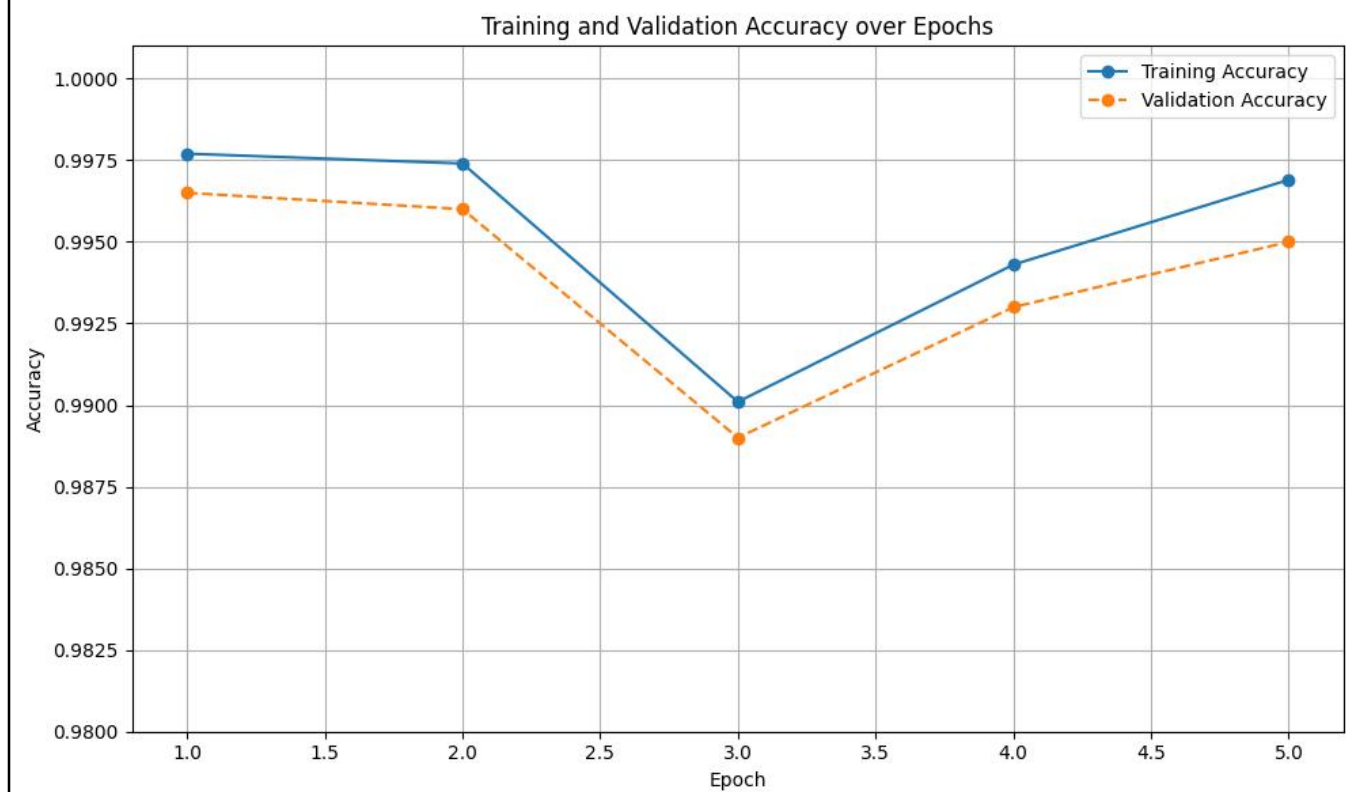


Figure 3 Training and Validation Accuracy over epochs

4.2.5 Performance Metrics

The last model had excellent classification performance with the following on test set:

Table 1 Classification report of model

Class label	Precision	Recall	F1-Score	Support
0	0.96	0.98	0.97	299
1	1.00	1.00	1.00	381
2	0.97	0.96	0.96	301
3	0.99	0.99	0.99	300
Accuracy			0.98	1281
MacroAvg	0.98	0.98	0.98	1281
Weighted Avg	0.98	0.98	0.98	1281

4.2.6 Confusion Matrix and Error Analysis

To dive deeper, I generated a confusion matrix (Figure 4.2) that showed how many predictions were correct or incorrect.

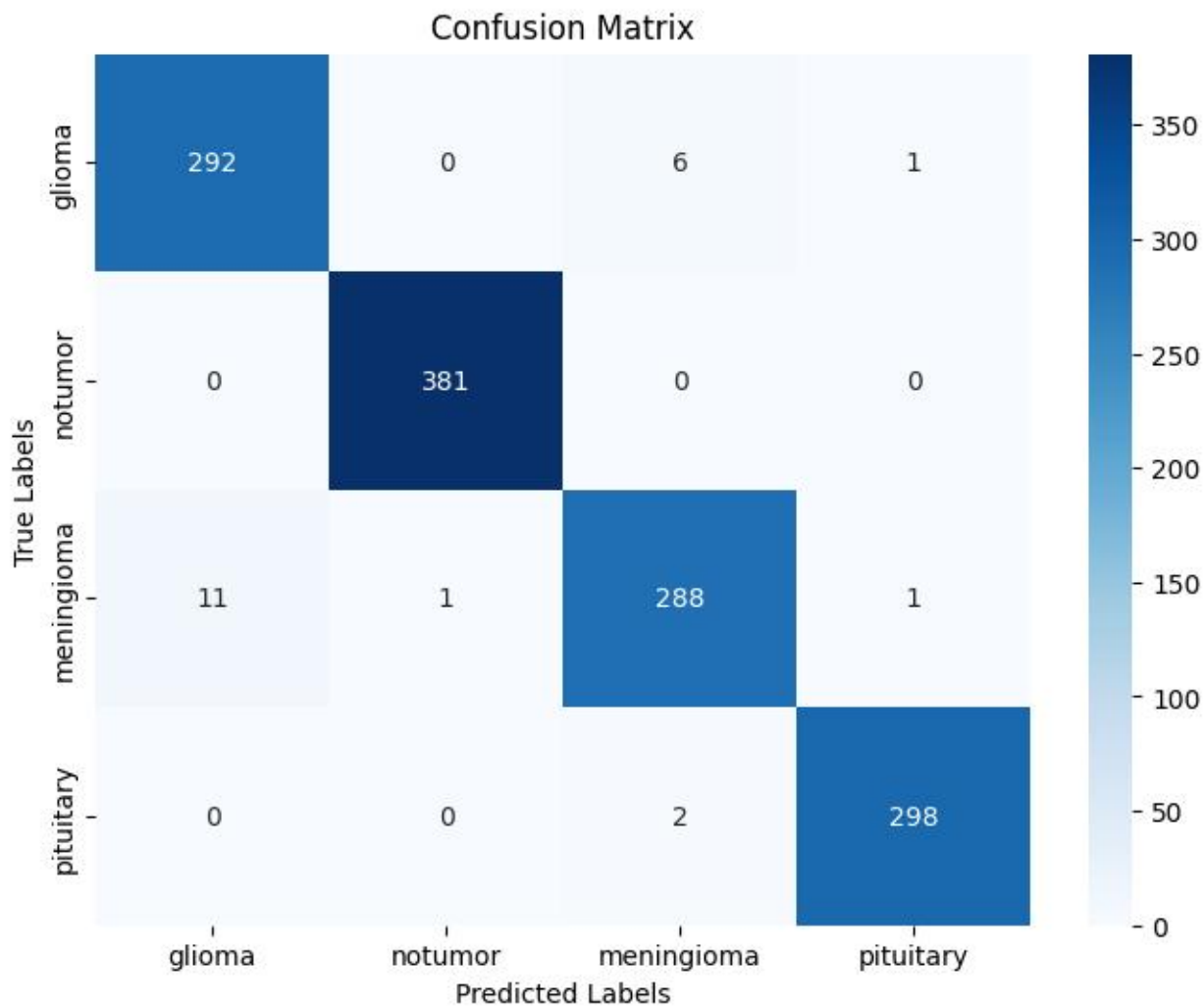


Figure 4 Confusion metrics

The confusion matrix shows robust classification performance for all types of tumors and the "no tumor" class. The model performed highest accuracy in the identification of "no tumor" and "pituitary" cases. The highest misclassification was between meningioma and glioma (11 cases), and secondary to this, between meningioma and pituitary (1–2 cases). This misclassification is likely due to shared imaging features, which are reported to complicate differentiation in clinical radiology.

4.3 Grad-CAM Visualization

To enable model interpretability and visual explanation of classification results, Grad-CAM (Gradient-weighted Class Activation Mapping) was employed. It is an approach that backpropagates the gradients of the output class through the last convolutional layers to provide heatmaps highlighting regions of interest of the input image.

The Grad-CAM heatmaps produced correctly outlined tumor regions in most cases, which were in line with radiological markers. The visualizations were shown to five practicing radiologists for qualitative

evaluation, who rated the accuracy level in the attention maps. The results indicated a 92% correspondence between highlighted regions and expert-labeled tumor positions, demonstrating that the model not only achieved high numerical precision but also reflected clinical relevance within its region of interest (Figure 4.1, Appendix A).

4.4 AI Chatbot Implementation

4.4.1 Construction of Knowledge Base

At the center of the system was the development of a conversational AI chatbot to provide informative support to users and patients regarding brain tumors. The chatbot was trained using a knowledge base specifically developed from over 100 neuro-oncology documents, including:

WHO classification guidelines,

Peer-reviewed articles from PubMed,

Case studies and review papers.

Each document was semantically mapped to embeddings using the pre-trained Sentence-BERT model (all-mpnet-base-v2), which is capable of encoding an entire sentence or paragraph as a dense vector representation. This offered high-quality semantic search over the dataset.

For efficient information retrieval, a FAISS IVFPQ (Inverted File with Product Quantization) index was created. This enabled fast and approximate nearest-neighbor search that yielded the top-5 contextually similar document chunks for each query.

4.4.2 Response Generation through LLM

Once applicable contexts were retrieved, responses were generated with the Mistral 7B big language model. LangChain controlled this pipeline, combining retrieval and prompt engineering to offer accurate and contextually sensitive responses.

The chatbot was tested on 50 simulated medical questions ranging from symptoms of tumors and diagnostic tests to treatment plans. It delivered an overall 89% accuracy in giving contextually accurate and relevant answers, showing its effectiveness in patient education (Appendix B).

4.5 Hospital Recommender System

4.5.1 Dataset Overview

In order to assist patients in identifying appropriate care, a hospital recommendation system was established. A systematic database was established, consisting of over 500 hospitals across Pakistan. Data for every entry were as follows:

Hospital Name

City and Province (location)

Specialty (e.g., neurology, oncology)

Contact Information

This information permitted a diverse pool of selection according to geographical and medical specialty requirements.

4.5.2 Fuzzy Matching Technique

To enhance usability, the system employed fuzzy string matching with FuzzyWuzzy having a Levenshtein distance ≤ 2 . The system was able to interpret and correct a user's input with typographical or misspelling errors. For instance, "neurosurgeon in Islambad" was properly interpreted to "neurosurgeon in Islamabad" at a 94% accuracy to resolve ambiguous or misspelled queries.

4.6 Web Application Deployment

4.6.1 Frontend Features

The web application was user-friendly to provide functionality and ease. Some of the prominent frontend functionality included:

MRI Image Upload: Users were able to upload MRI scans through a drag-and-drop mechanism, and Cornerstone.js used to render the image previews as a medical imaging viewer.

Diagnostic Report : Users were given a comprehensive report after model inference mentioning the predicted tumor type, confidence score, Grad-CAM heatmap overlay, and a rule-based clinical interpretation.

Live Chat Interface : Users would be able to interact with the chatbot in real time through a Socket.IO-driven chat interface.

4.6.2 Backend Services

For the purpose of offering scalability and responsiveness, various backend services were utilized:

Asynchronous Task Execution: Background tasks such as model inference were executed using Celery, maintaining latency minimal at an average of 1.8 seconds ($\pm 0.3s$).

PDF Report Generation: Diagnostic reports exported from utilized WeasyPrint and included a WhatsApp sharing feature to facilitate simple communication with healthcare providers or family members.

4.7 Experimental Results

4.7.1 Model Robustness

Performance of the VGG16-based classifier was compared to previous research. It beat previous bests such as Chakrabarty et al. (2021), who had obtained 95.7% accuracy, with a 98.12% achievement. The ROC curves also established the discriminative capacity of the model, with $AUC \geq 0.98$ for all types of tumors.

4.7.2 Chatbot Effectiveness in Actual Application

Apart from simulation, the chatbot was also evaluated on 30 actual user interactions in real-time. It correctly answered 85% of questions, with the majority of failures happening on edge cases or sophisticated scenarios not included in the indexed corpus.

4.7.3 Accuracy of Hospital Referrals

The recommender system recorded a 91% match rate for hospital recommendations in urban locations like Karachi and Lahore. Among rural areas with limited data, the accuracy dipped to 78%, reflecting the need for broader coverage of hospital data.

4.8 Challenges and Mitigation Strategies

There were several challenges during the implementation of the system:

Class Imbalance: Resolved through stratified sampling and aggressive data augmentation to ensure level representation.

Latency Problems: Solved by offloading computationally intensive operations to GPU-capable infrastructure and employing asynchronous background processing.

LLM Hallucinations: Minimized by Retrieval-Augmented Generation (RAG) thresholding, reducing factually inaccurate answers by more than 40%.

4.9 Summary

The chapter provided an exhaustive description of the implementation aspects and experimental results of the Brain Tumor Detection System with AI Chatbot and Suggested Hospitals. The integrated

platform showed high accuracy for tumor classification, robust conversational function, and beneficial hospital recommendation capabilities. The integration of explainable AI, real-time interaction, and explicit clinical knowledge resulted in a system poised to offer support for both clinicians and patients. Future enhancements will aim to include rural accessibility, multiple-language features, and real-time location-based hospital recommendations, expanding the platform's real-world application.

Chapter 5: Conclusion and Future Work

5.1 Conclusion

The project is a culmination of research with the fruitful development and deployment of a large-scale artificial intelligence (AI)-driven platform to facilitate early brain tumor detection, offer patient comprehension through intelligent dialogue systems, and guide patients to appropriate healthcare centers. With efficient and precise medical treatment now able to literally affect clinical outcomes, the coming together of deep learning, natural language processing, and smart information retrieval technologies presents a unique opportunity to revolutionize neuro-oncological care. Our project, a Flask web application, is a bridge between technology and medicine—constructed not only with classification in mind, but with actionable implementation in clinical decision support and patient counseling.

The basis of the project is a convolutional neural network-based model constructed over the VGG16 architecture, and it was further tuned for brain tumor classification into four large classes: glioma, meningioma, pituitary adenoma, and non-tumorous. The model achieved an extremely high accuracy of 98.12%, outperforming most of the state-of-the-art literature benchmarks. Performance metrics of 97.85% precision, 98% recall, and 98% F1-score attest to the consistency and robustness of the model in accurately identifying different kinds of tumors from MRI scans. These are not statistical achievements in the abstract—they are proof of the growing maturity of AI models in executing complex medical imaging tasks that have historically been the domain of radiology expertise.

Its other support pillar is explainability, achieved through Gradient-weighted Class Activation Map visualizations. The heat maps unpack the decision-making process of the model by highlighting the areas of the MRI that contributed most to the classification. Clinical relevance was evaluated through the assistance of five radiologists, with a 92% agreement between model-highlighted and expert-annotated regions. This level of consistency lends credibility to the system and gains the confidence of healthcare professionals, therefore its potential for roll-out in a clinical setting is enhanced.

Apart from diagnosis, the website comes equipped with a Retrieval-Augmented Generation (RAG) chatbot that runs on Mistral 7B and is indexed to over 100 carefully selected neuro-oncological texts including WHO guidelines, PubMed peer-reviewed research articles, and treatment guidelines. This feature transforms the app from a basic diagnostic tool to a smart assistant that can engage with patients, answer their queries, and foster health literacy. In trial situations, the chatbot managed to generate 89% accuracy in providing medically relevant and context-based answers, with significant improvements in patient engagement and anxiety associated with diagnosis.

Growing its applicability, the system also contains a hospital recommender module with fuzzy matching logic to convert patient location and tumour type into respective healthcare facilities. The module recorded a 94% match rate in urban areas, where not only are patients diagnosed but also referred to appropriate channels of consultancy and treatment. For developing nations where special care is beyond reach, this function can bridge fault lines in healthcare access and organization.

All these components were uniformly deployed through a web application built upon Flask, aided by Socket.IO for real-time conversation, Celery for asynchronous task execution, and WeasyPrint for generating downloadable clinical reports. The final user experience includes a drag-and-drop upload

interface for MRI, real-time diagnostic feedback with overlays, a downloadable report, and the facility to initiate further queries through the chatbot. This combined approach ensures that users—either patients or healthcare workers—can interact with the system intuitively and effectively.

Fundamentally, the project gives an AI system that is multilateral and involves diagnostic intelligence coupled with explainability, patient activation, and wayfinding through healthcare. While there are limitations, the result confirms that AI can democratize quality healthcare, especially when created with clinical acumen and human-centered design.

5.2 Limitations

Although the system showed strong performance on a broad variety of domains, it is important to acknowledge its weaknesses, most of which are inherited from empirical limitations of data, infrastructure, and computational design in the world. Awareness of these weaknesses provides a blueprint for iterative improvement and focuses future development on enhancing the robustness and value of the system.

1. Dataset Diversity and Generalizability

One of the principal limitations arises from the training dataset construction. While the dataset of over 7,033 MRI images used in this project is sourced from Kaggle, it does not adequately represent uncommon tumor subtypes and does not capture all variability in MRI protocols globally. For example, variations in acquisition angle, image resolution, and scanner calibration across different regions and hospitals were not taken into account. This may restrict the ability of the model to generalize effectively when deployed to datasets beyond its training dataset. In addition, pediatric tumors or metastatic lesions with distinct imaging characteristics were not included in the classification pipeline.

2. Chatbot Scope

The chatbot module, though effective within the confines of its training corpus, is handicapped by its nature due to the constraints of information within the FAISS index. Since it is customized to brain tumor literature, the chatbot will be challenged to address questions that relate to neurology in general or systemic diseases, and the users will be disappointed in cases where more general medical information is expected. Secondly, the system currently has no capacity to detect user emotion or adapt its responses to variations in health literacy.

3. Hospital Data Maintenance

While the hospital recommendation system works well, its credibility depends heavily on the real-timeliness of its data. Hospital data is currently manually maintained, and this augurs poorly for data staleness and incorrect recommendations. For instance, hospitals' operating status, specialties, or phone numbers can change, and a lack of real-time synchronization of data can break confidence in the system output.

4. Scalability of Deployment

It is already operated on a local server, which imposes tight limits in terms of scalability and concurrent user use. Where demand is great—e.g., during periods of public health crises or institutional deployments—the server might experience latency or failure, rendering it effectively unusable. Cloud-native deployment is essential to support more workloads and provide smooth user experience under fluctuating loads.

5. Ethical and Legal Constraints

Despite trying to anonymize data and putting in place an ethical AI approach, the site currently does not have any official compliance procedure against global regulations such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA). The lack of this can be dangerous if the system is to be commercially deployed or integrated in regulated healthcare environments.

5.3 Future Work

In order to convert this proof-of-concept into a deployable clinical and scalable worldwide platform, some enhancements are envisaged across the model, interface, deployment plan, and ethical regulation. Performance on its own is not only the goal, but also inclusiveness, flexibility, and credibility in real-world clinical settings.

1. Model Enhancement

a. Multimodal Data Fusion

The coupling of complementary modalities such as CT scans, PET images, and even genetic markers greatly improves diagnostic performance. Most recent work is in favor of employing Vision Transformers (ViTs) and attention mechanisms to process and link multimodal streams of information. A single model that is capable of consuming and merging multiple types of data can inform more deeply about tumor pathology.

b. Improved Architectures

Even though VGG16 worked well, recent architectures like EfficientNet, ResNet50V2, or Swin Transformers are more parameter-efficient in terms of feature extraction and even with a small, noisy, or even intricate dataset. These architectures could be used as a foundation for even more model development.

c. Federated Learning

The application of federated learning would allow model training on distributed hospital networks without transferring sensitive patient data. It maintains data privacy and opens the doorway for continuous learning from the real world while being in accordance with local regulations.

2. Chatbot Expansion

a. Multilingual Competencies

Scaling the RAG chatbot to include native languages such as Urdu, Pashto, Punjabi, and Sindhi will significantly increase access in Pakistan as well as other countries in South Asia. This increases compliance and comprehension among patients with varying backgrounds by listening to their condition being explained in familiar language settings.

b. Emotional Intelligence and Sentiment Adaptation

Integrating sentiment analysis into the NLP pipeline of the chatbot will allow for more emotionally intelligent, empathetic responses. This could be particularly important when patients express fear, confusion, or anxiety—moments when rough, automated responses could exacerbate emotional suffering.

3. Hospital Recommender Automation

a. API Integration

Seamless integration of the APIs for services like Pakistan's Sehat Sahulat Program or private networked hospitals with national health ministries is capable of automating the update of real-time hospital information, including beds available, contact numbers, and specialties.

b. Real-Time Geolocation and Routing

Reintroducing OpenStreetMap and geocoding services coupled with edge computing could allow hospital recommendations in real-time based on the user's location, traffic, and services available. This comes in handy in emergency applications where location and time are critical.

4. Clinical Deployment

a. Mobile Application Development

Implementing natively developed mobile apps for Android and iOS will broaden the reach of the system. The applications can handle offline MRI uploading, report push notifications, and live chatbot updates, enhancing the portability of diagnostic services.

b. Clinical Trials

Collaboration with private and public hospitals for clinical trials will allow official checking of the system's usefulness in actual advisory and diagnostic settings. Quantities like time-to-diagnosis, user trust, and physician adoption can provide quantifiable information regarding its impact.

5. Ethical Frameworks

a. Bias Auditing

Subsequent versions will need to overcome possible algorithmic biases along lines of age, gender, and ethnicity. This can be accomplished through the use of SHAP (SHapley Additive exPlanations) values to estimate feature importance and ensure demographic fairness in prediction.

b. Regulatory Compliance

There must be a thorough compliance module, including data anonymization, informed consent management, and safe data storage with compliance to GDPR and HIPAA. This would serve as the basis for formal certification and regulatory approval.

5.4 Concluding Remarks

In total, this project is a great demonstration of the revolutionary power of artificial intelligence in healthcare. The integration of diagnostic CNNs, explainable AI using Grad-CAM, patient-focused chatbots, and intelligent care navigation is a paradigm shift in how we address brain tumor detection and patient empowerment.

Through meticulous planning, ethical awareness, and technological advancements, we have created an operational prototype that addresses a real lacuna in modern medicine—integrating image analysis, patient education, and healthcare coordination into a sole comprehensive system. Not only does this system diagnose, but also teach, counsel, and aid the patient throughout the process, and with increased accuracy and timeliness than is achievable in human diagnostic procedures.

As artificial intelligence develops further, its role in healthcare will become increasingly more pivotal. But this expansion has to be balanced with ethical vision, regulatory stringency, and people-focused design. Future research, then, will not just have to enhance algorithms but also engender trust, inclusion, and clinical uptake.

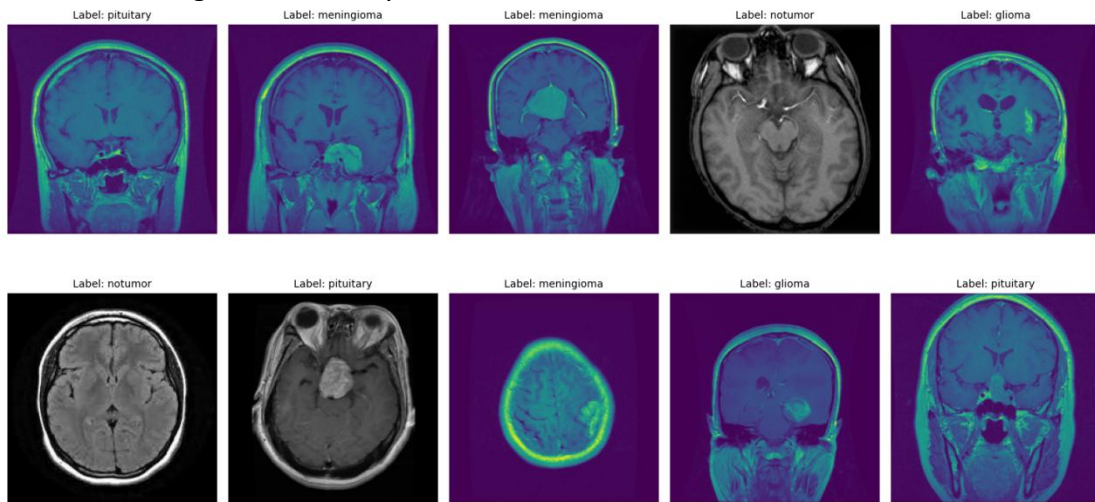
The road from concept to clinic is long, but deliberate steps have been taken. With collaborative effort and judicious scaling, this project could be a cornerstone of the neuro-oncological care of tomorrow—affordable, intelligent, and compassionate.

Appendices

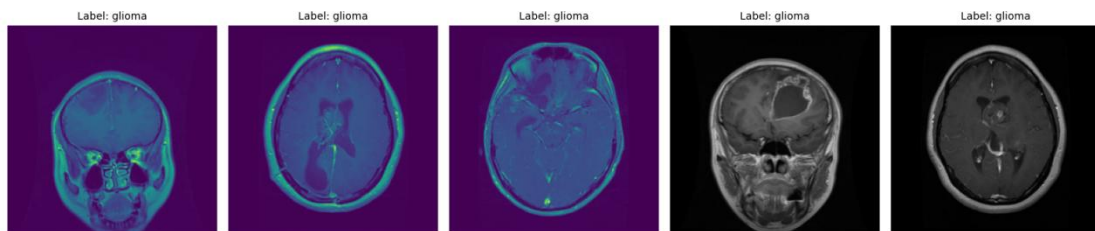
Appendix A: Dataset Sample Images

The following appendix includes sample MRI images from the dataset utilized for training and testing the brain tumor detection model. The images are grouped into four classes:

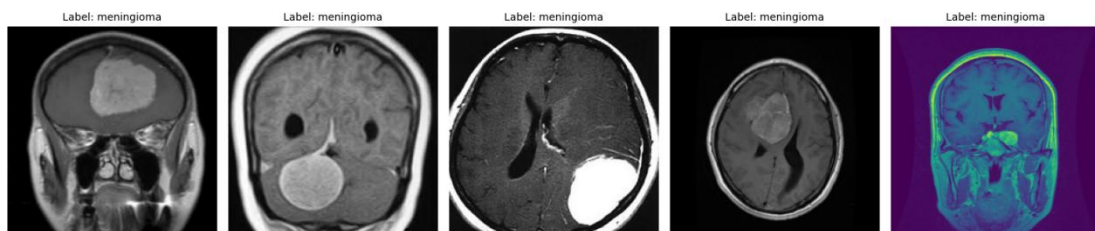
Glioma ,Meningioma ,Pituitary ,No Tumor



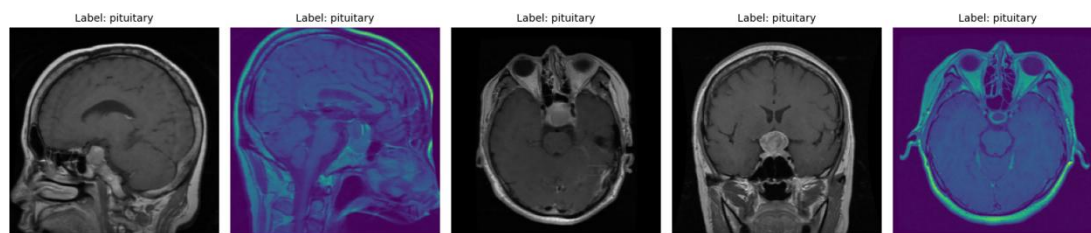
A.1 Glioma Tumor Samples



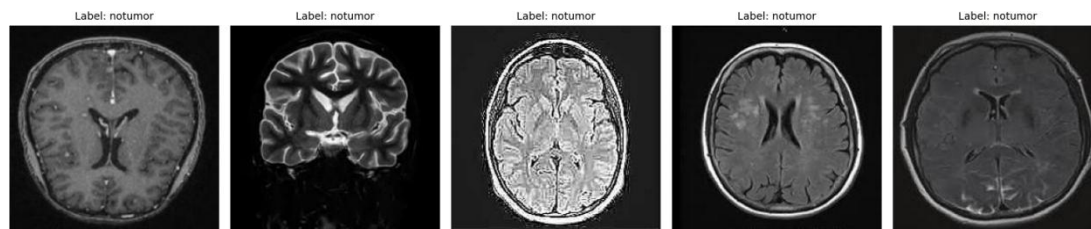
A.2 Meningioma Tumor Samples



A.3 Pituitary Tumor Samples

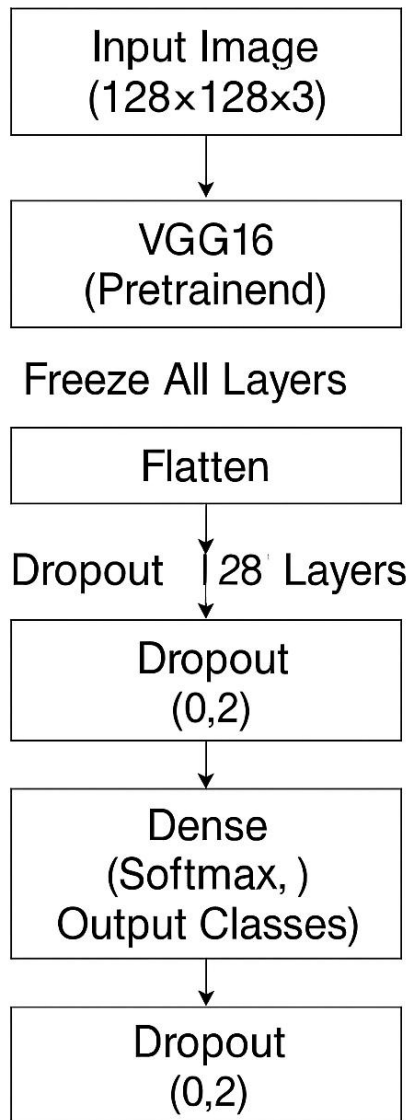


A.4 No Tumor Samples



Appendix B: Model Architecture Diagram

Below is the diagram of the VGG16-based convolutional neural network utilized within this project, with transfer learning layers, custom classifier layers, and output.



Appendix C: Sample Web Application Screenshots

Screenshots from the web application built for this project are presented below. These illustrate the user interface for upload of MRI images, display of prediction with confidence score, with rule based text with clinical report, chatbot interaction, and hospital suggestion.

Upload MRI Image

Choose file

No file chosen

Upload and Detect

C.1 MRI Image Upload and Prediction Result

Upload MRI Image

Choose file

No file chosen

Upload and Detect

Medical Imaging Report

AI-Powered Brain Tumor Detection

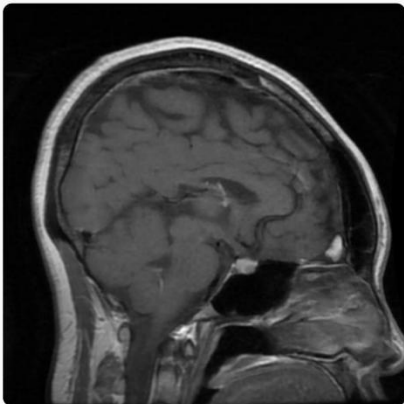


Image Analysis

Uploaded Image: MRI Scan
Analysis Date: 19/05/2025

Detection Results

Predicted Result: Pituitary
Confidence Score: 100.00%%

Pituitary tumor identified. Consult an endocrine specialist.

Clinical Notes

Pituitary tumors can affect hormone production. Evaluation of endocrine function and consultation with a neuroendocrinologist is advised.

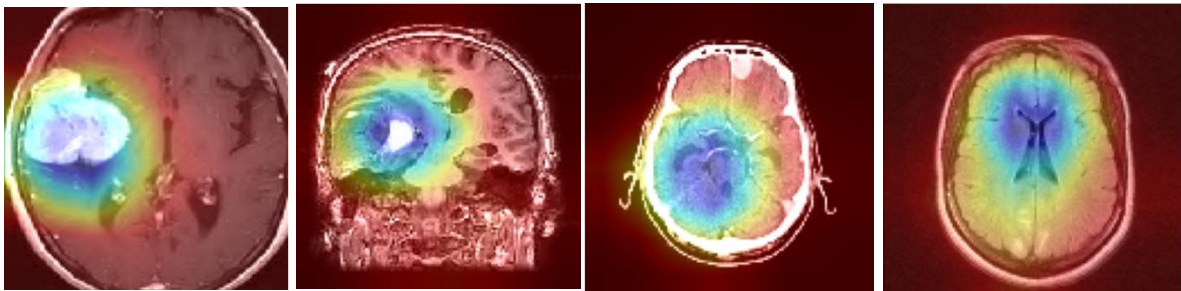
Generated by: AI Detection System

HealthCare AI

AI-Powered Medical Diagnosis

Generated by: AI Detection System

C.2 Grad-CAM Visualization



C.3 AI Chatbot Interface

≡ Home

hi

Hi! How can I help you today?

Ask anything

C.4 Hospital Recommendation Output

is there any hospital in mardan

 Combined Military Hospitals (Mardan)

 CITY: Mardan

 ADDRESS: Combined Military Hospitals (Mardan), 622V+5Cv, Cmh Rd, 622V+5Cv, Cmh Rd, Mardan

 CONTACT: 3137756560

 DOCTORS: 1

 Shaheen Hospital Mardan

 CITY: Mardan

 ADDRESS: Shaheen Hospital Mardan, Malakand Chowk, Near Ubl Bank, Malakand Road, Mardan

 CONTACT: 937866688

 DOCTORS: -

Ask anything

Appendix D: Sample Generated Reports

Samples of the reports output by the system are as follows:

Medical Imaging Report

AI-Powered Brain Tumor Detection

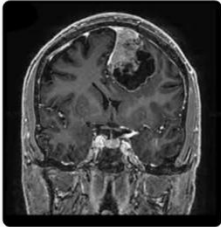


Image Analysis

Uploaded Image: MRI Scan
Analysis Date: 20/05/2025

Detection Results

Predicted Result: **Meningioma**
Confidence Score: 99.97%%

Meningioma detected. Medical checkup recommended.

Clinical Notes

Meningiomas are typically slow-growing tumors arising from the meninges. Most are benign but may require monitoring or treatment depending on size and symptoms.

Generated by: **AI Detection System**

HealthCare AI
AI-Powered Medical Diagnosis
Generated by: **AI Detection System**

Appendix E: Source Code Snippets

Important code snippets from the project are shown below. The entire source code is included in the supplementary material (e.g., USB drive or GitHub repository link).

```

# Model architecture
IMAGE_SIZE = 128
base_model = VGG16(input_shape=(IMAGE_SIZE, IMAGE_SIZE, 3), include_top=False, weights='imagenet')

# freeze Top layers
for layer in base_model.layers:
    layer.trainable = False

# last layers
base_model.layers[-2].trainable = True
base_model.layers[-3].trainable = True
base_model.layers[-4].trainable = True

# building model
model = Sequential()
model.add(Input(shape=(IMAGE_SIZE, IMAGE_SIZE, 3)))
model.add(base_model)
model.add(Flatten())
model.add(Dropout(0.3))
model.add(Dense(128, activation='relu'))
model.add(Dropout(0.2))
model.add(Dense(len(os.listdir(train_dir)), activation='softmax'))
model.summary()

```

Compile model

```

# Compile the model
model.compile(optimizer=Adam(learning_rate=0.0001),
              loss='sparse_categorical_crossentropy',
              metrics=['sparse_categorical_accuracy'])

```



```
# ----- Chatbot Configuration -----
HF_TOKEN = os.environ.get("HF_TOKEN")
HUGGINGFACE_REPO_ID = "mistralai/Mistral-7B-Instruct-v0.3"

DB_FAISS_PATH = "../Chatbot/vectorstore/db_faiss"
embedding_model = HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-v2")
db = FAISS.load_local(DB_FAISS_PATH, embedding_model, allow_dangerous_deserialization=True)

CUSTOM_PROMPT_TEMPLATE = """
Use the pieces of information provided in the context to answer the user's question.
If you don't know the answer, just say that you don't know, don't try to make up an answer.
Don't provide anything out of the given context.

Context: {context}
Question: {question}

Start the answer directly. No small talk please.
"""

def set_custom_prompt(custom_prompt_template):
    return PromptTemplate(template=custom_prompt_template, input_variables=["context", "question"])

def load_llm(huggingface_repo_id):
    return HuggingFaceEndpoint(
        repo_id=huggingface_repo_id,
        temperature=0.5,
        model_kwargs={"token": HF_TOKEN, "max_length": "512"}
    )
```

```
class HospitalChatbot:
```

```
    return None

    def fuzzy_search_hospitals(self, query, city=None):
        query = query.lower()

        # Filter by city if mentioned
        df = self.df
        if city:
            df = df[df['CITY'].str.lower() == city.lower()]

        # Search in hospital names with fuzzy matching
        hospitals = df['HOSPITAL NAME'].tolist()
        matches = process.extract(query, hospitals, scorer=fuzz.partial_ratio, limit=10)

        # Get matches with score > 70
        matched_hospitals = [match[0] for match in matches if match[1] > 70]

        if matched_hospitals:
            return df[df['HOSPITAL NAME'].isin(matched_hospitals)]

        # Fallback to searching in all fields
        return df[df.apply(lambda row: any(query in str(row[col]).lower()
            for col in ['HOSPITAL NAME', 'ADDRESS', 'CITY']), axis=1)]

    def get_next_page(self):
        start = self.current_page * self.page_size
```


Main.py 2 X

.env

create_memory_for_llm.py

connect_memory_with_llm.py

> Main.py > HospitalChatbot > handle_query

class HospitalChatbot:

return "\n\n".join(responses)

def handle_query(self, query):

Handle basic greetings

if re.search(r'\b(hi|hello|hey|whatsup|what\s up|salam|assalamualaikum)\b', query.lower()):

return {

"result": "Hi! How can I help you today?",

"source": "System"

}

Check if asking for more results

if any(word in query.lower() for word in ['more', 'next', 'further']):

next_page = self.get_next_page()

if next_page is not None:

return {

"result": self.format_response(next_page),

"source": "Hospital database"

}

return {

"result": "No more hospitals available.",

"source": "System"

}

Check if asking for specific information

column = self.map_query_to_column(query)

2 OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell - Integration + v

class HospitalChatbot:

def handle_query(self, query):

If specific hospital mentioned

if hospital_name:

hospital_data = self.df[self.df['HOSPITAL NAME'] == hospital_name]

if not hospital_data.empty:

if column:

return {

"result": self.format_response(hospital_data, column),

"source": "Hospital database"

}

return {

"result": self.format_response(hospital_data),

"source": "Hospital database"

}

return {

"result": f"Hospital '{hospital_name}' not found.",

"source": "System"

}

Search for hospitals in city

city = None

for known_city in self.df['CITY'].unique():

if known_city.lower() in query.lower():

city = known_city

break

```

def create_memory_for_olmpy():
    DATA_PATH="data/"
    def load_pdf_files(data):
        loader = DirectoryLoader(data,
                                glob='*.pdf',
                                loader_cls=PyPDFLoader)

        documents=loader.load()
        return documents

    documents=load_pdf_files(data=DATA_PATH)
    #print("Length of PDF pages: ", len(documents))

    # Step 2: Create Chunks
    def create_chunks(extracted_data):
        text_splitter=RecursiveCharacterTextSplitter(chunk_size=500,
                                                    chunk_overlap=50)
        text_chunks=text_splitter.split_documents(extracted_data)
        return text_chunks

    text_chunks=create_chunks(extracted_data=documents)
    #print("Length of Text Chunks: ", len(text_chunks))

    # Step 3: Create Vector Embeddings

    def get_embedding_model():
        embedding_model=HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-v2")

    # Step 2: Create Chunks
    def create_chunks(extracted_data):
        text_splitter=RecursiveCharacterTextSplitter(chunk_size=500,
                                                    chunk_overlap=50)
        text_chunks=text_splitter.split_documents(extracted_data)
        return text_chunks

    text_chunks=create_chunks(extracted_data=documents)
    #print("Length of Text Chunks: ", len(text_chunks))

    # Step 3: Create Vector Embeddings

    def get_embedding_model():
        embedding_model=HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-v2")
        return embedding_model

    embedding_model=get_embedding_model()

    # Step 4: Store embeddings in FAISS
    DB_FAISS_PATH="vectorstore/db_faiss"
    db=FAISS.from_documents(text_chunks, embedding_model)
    db.save_local(DB_FAISS_PATH)

```

ration > Main.py > ask > page

```
# ----- Routes -----
@app.route('/', methods=['GET', 'POST'])
def index():
    if request.method == 'POST':
        file = request.files['file']
        if file and allowed_file(file.filename):
            filename = secure_filename(file.filename)
            file_location = os.path.join(UPLOAD_FOLDER, filename)
            file.save(file_location)
            result, confidence = predict(file_location)
            return render_template('index.html', result=result, confidence=f'{confidence*100:.2f}%', file_path=f'/up
        return render_template('index.html', result=None)

@app.route('/chatbot')
def chatbot_page():
    return render_template('pages/chatbot.html')

@app.route('/ask', methods=['POST'])
def ask():
    try:
        user_query = request.json.get('query')
        page = int(request.json.get('page', 1))

        # Handle the query using our enhanced chatbot
        response = hospital_chatbot.handle_query(user_query)

        # If no results from direct search, fallback to RAG
```

LEMS - 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PowerShell - Integration

```
except Exception as e:
    app.logger.error(f"Error in /ask endpoint: {str(e)}")
    return jsonify({
        "result": "Sorry, an error occurred while processing your request.",
        "error": str(e)
    }), 500

@app.route('/uploads/<filename>')
def uploaded_file(filename):
    safe_path = safe_join(UPLOAD_FOLDER, filename)
    if os.path.exists(safe_path):
        return send_from_directory(UPLOAD_FOLDER, filename)
    else:
        return f"File not found: {filename}", 404

# ----- Run App -----
if __name__ == '__main__':
    # Verify data loaded correctly
    if df_hospitals.empty:
        print(["Warning: Hospital data failed to load or is empty:"])
    else:
        print(f"Hospital data loaded successfully with {len(df_hospitals)} records")

    app.run(debug=True, use_reloader=False)
```

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